prompt与实验内容对照-LCAD_RASA_latest_experimental_section_mapping

Experimental Section and Execution-Prompt Mapping for the SCI Paper

Paper Theme

LLM-Augmented Cross-Modal Semantic Alignment for Large-Scale Multicentre Cervical Analytics

Method:

LCAD-RASA: Label-Constrained Agent Distillation and Report-Anchored Semantic Alignment

This document maps the final execution prompts to the experimental section of the SCI paper. It can be used to organise the experiments, write the Methods/Experiments section, and track which code module supports each claim.

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1. Experimental Overview

The experiments are designed to verify the following scientific question:

> Can LLM-augmented label-constrained distillation convert weak binary labels into reliable report-level semantic supervision, and can report-anchored semantic alignment improve cross-modal representation learning across OCT, colposcopy and clinical instruction data in large-scale multicentre cervical analytics?

The experimental logic follows a reviewer-proof chain:

代码块

```
Data problem exists
→ LCAD pseudo reports are calibrated
→ LCAD is not label-template expansion
→ pseudo-report QC is necessary
→ RASA improves over simple fusion
→ the model uses multimodal evidence
```

```
→ the method generalises across centres
→ report-missing centres have clinically plausible outputs
```

---

2. Experiment-to-Prompt Mapping

Experiment	Paper Purpose	Execution Prompt / Script	Main Output
Exp 0	Build project and reproducible pipeline	Project creation prompt	Project structure, README, configs
Exp 1	Prove multicentre semantic supervision imbalance	`00_audit_data.py`, `01_build_manifest.py`	Data audit, manifest, centre summary
Exp 2	Extract modality evidence for Agent and model	`02_extract_modality_evidence.py`	OCT, colposcopy, instruction evidence files
Exp 3	Calibrate LCAD on report-rich centre	`03_report_rich_masking_validation.py`	Masking validation metrics
Exp 4	Generate structured pseudo reports	`04_generate_pseudo_reports.py`	Pseudo reports for report-missing centres
Exp 5	Filter pseudo reports and assign weights	`05_qc_pseudo_reports.py`	QC table, weighted manifest
Exp 6	Train baseline and full LCAD-RASA	`06_train_lcad_rasa.py`	Model checkpoints
Exp 7	Evaluate report quality and clinical consistency	`07_eval_lcad_rasa.py`	Evaluation tables
Exp 8	Run ablations	`08_run_ablation.py`	Ablation summaries
Exp 9	Test cross-centre generalisation	LOCO inside `08_run_ablation.py`	Leave-one-centre-out table
Exp 10	Export physician review package	`10_export_physician_review_pack.py`	Review cases and scoring form
Exp 11	Export paper-ready results	`11_export_results.py`	Main and supplementary tables

3. Experiment 1: Multicentre Semantic Supervision Profiling

Aim

To establish the real-world data problem addressed by the paper: cross-centre report-level semantic supervision imbalance.

Execution

代码块

```
python scripts/00_audit_data.py \
  --data_root /data \
  --output outputs/data_audit_report.md

python scripts/01_build_manifest.py \
  --data_root /data \
  --output outputs/manifests/full_manifest.csv
```

What This Experiment Verifies

This experiment verifies that:

1. all five centres contain multimodal cervical data;
2. OCT, colposcopy and clinical instruction are available at scale;
3. only one centre contains real diagnostic reports;
4. the other four centres contain binary labels but lack reports;
5. the binary label endpoint must be defined or manually confirmed.

Metrics and Outputs

代码块

```
Number of centres
Number of cases per centre
OCT availability per centre
Colposcopy availability per centre
Instruction availability per centre
Real report availability per centre
Binary label availability per centre
Missing modality rate
```

```
Binary label endpoint definition
```

Outputs:

代码块

```
outputs/data_audit_report.md
outputs/tables/centre_modality_summary.csv
outputs/tables/report_availability_summary.csv
outputs/tables/binary_label_endpoint_definition.csv
outputs/manifests/full_manifest.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.1 Dataset and Semantic Supervision Profiling
```

Reviewer Question Addressed

> Is the proposed problem real, or artificially created?

This experiment demonstrates that the study addresses a genuine multicentre semantic supervision imbalance.

---

4. Experiment 2: Modality Evidence Extraction

Aim

To extract modality-level evidence used by LCAD and RASA.

Execution

代码块

```
python scripts/02_extract_modality_evidence.py \
  --manifest outputs/manifests/full_manifest.csv \
  --output_dir outputs/modality_evidence
```

What This Experiment Verifies

This experiment ensures that pseudo reports are not generated solely from binary labels. It provides case-level evidence from:

1. OCT;
2. colposcopy;
3. clinical instruction.

Key Requirement

Formal LCAD experiments should not rely only on metadata such as image number or image size. Metadata-only summaries can be used for audit and debugging but should be flagged as insufficient for modality-grounded pseudo-report generation.

Outputs

代码块

```
outputs/modality_evidence/{center_id}/{case_id}.json
outputs/tables/modality_evidence_status.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.2 Modality Evidence Preparation
```

Reviewer Question Addressed

> Are pseudo reports grounded in actual modality evidence, or generated from labels alone?

---

5. Experiment 3: Report-Rich Centre Masking Validation

Aim

To calibrate the LCAD pseudo-report generation strategy using the only centre that contains real reports.

Execution

代码块

```
python scripts/03_report_rich_masking_validation.py \
```

```
    --config configs/distill.yaml \
    --manifest outputs/manifests/full_manifest.csv \
    --evidence_dir outputs/modality_evidence \
    --client mock
```

Design

In the report-rich centre:

1. hide the real report;
2. provide the Agent with available evidence;
3. generate a pseudo report;
4. compare the pseudo report with the hidden real report.

Three Agent settings are compared:

代码块

```
Label-only Agent
Modality-only Agent
Modality-plus-label Agent
```

The formal LCAD uses modality-plus-label.

Metrics

代码块

```
BLEU
ROUGE-L
BERTScore
Section completeness
Impression consistency
Label consistency
Contradiction rate
Hallucinated diagnosis rate
Modality-missing hallucination rate
```

Outputs

代码块

```
outputs/tables/report_rich_masking_validation_metrics.csv
outputs/tables/report_rich_masking_validation_cases.csv
```

```
outputs/masking_validation/masked_pseudo_reports/
```

Corresponding Paper Section

代码块

```
Experiments 4.3 Calibration of Label-Constrained Agent Distillation
```

Reviewer Questions Addressed

> Are pseudo reports trustworthy?
> Are pseudo reports only label templates?
> Does modality-plus-label generation outperform label-only generation?

Paper Interpretation

This experiment calibrates LCAD; it does not claim that report-missing centre pseudo reports are ground truth.

---

6. Experiment 4: LCAD Pseudo-Report Generation

Aim

To generate structured pseudo reports for the four report-missing weak-label centres.

Execution

代码块

```
python scripts/04_generate_pseudo_reports.py \
  --config configs/distill.yaml \
  --manifest outputs/manifests/full_manifest.csv \
  --evidence_dir outputs/modality_evidence \
  --client mock \
  --setting modality_plus_label
```

What This Experiment Verifies

This experiment tests whether binary labels can be expanded into structured, evidence-attributed weak report-level supervision under modality and label constraints.

Required Output Sections

代码块

```
diagnostic_summary
oct_findings
colposcopy_findings
clinical_context
impression
recommendation
sentence_level_evidence
label_consistency
confidence
quality_flags
```

Outputs

代码块

```
outputs/pseudo_reports/{center_id}/{case_id}.json
outputs/tables/pseudo_report_generation_status.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.4 LCAD Pseudo-Report Generation for Report-Missing Centres
```

Reviewer Question Addressed

> How are the report-missing centres incorporated into report-level learning?

---

7. Experiment 5: Pseudo-Report QC and Reliability Weighting

Aim

To filter noisy LLM/Agent-generated pseudo reports and assign reliability weights.

Execution

代码块

```
python scripts/05_qc_pseudo_reports.py \
  --manifest outputs/manifests/full_manifest.csv \
  --pseudo_report_dir outputs/pseudo_reports \
```

```
  --output_manifest outputs/manifests/full_manifest_with_pseudo_reports.csv
```

QC Criteria

代码块

```
JSON validity
Required section completeness
Label-impression consistency
Positive/negative contradiction
Pathological hallucination
Modality-missing hallucination
Sentence-level evidence validity
Confidence validity
Privacy leakage
Report length and templating
```

Reliability Weight

代码块

```
pseudo_training_weight = confidence × qc_score
```

Outputs

代码块

```
outputs/qc/pseudo_report_qc_cases.csv
outputs/tables/pseudo_report_quality_summary.csv
outputs/manifests/full_manifest_with_pseudo_reports.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.5 Pseudo-Report Quality Control and Reliability Weighting
```

Reviewer Question Addressed

> How do you prevent noisy Agent outputs from corrupting the training signal?

---

8. Experiment 6: Real-Only Baseline

Aim

To establish the baseline performance when only the report-rich centre is used.

Execution

代码块

```
python scripts/06_train_lcad_rasa.py \
  --config configs/train.yaml \
  --experiment real_only
```

Training Data

代码块

```
Real reports from the report-rich centre only
```

Evaluation

代码块

```
python scripts/07_eval_lcad_rasa.py \
  --config configs/eval.yaml \
  --checkpoint outputs/checkpoints/real_only/best.ckpt \
  --manifest outputs/manifests/full_manifest_with_pseudo_reports.csv \
  --experiment real_only
```

Metrics

For real-report cases:

代码块

```
BLEU
ROUGE-L
BERTScore
Section completeness
Clinical consistency
```

Corresponding Paper Section

代码块

```
Experiments 4.6 Baseline with Real Reports Only
```

Reviewer Question Addressed

> What is the performance if the model is trained only on available real reports?

---

9. Experiment 7: Pseudo-Augmented Training

Aim

To test whether QC-passed pseudo reports improve model training beyond the real-only baseline.

Execution

代码块

```
python scripts/06_train_lcad_rasa.py \
  --config configs/train.yaml \
  --experiment pseudo_augmented
```

Training Data

代码块

```
Real reports from the report-rich centre
+ QC-passed pseudo reports from report-missing centres
```

Key Comparison

代码块

```
real_only vs pseudo_augmented
```

Metrics

代码块

```
Reference-based metrics on real-report test set
```

```
Clinical consistency on all centres
Section completeness
Hallucination rate
Per-centre stability
```

Corresponding Paper Section

代码块

```
Experiments 4.7 Effect of LCAD-Augmented Report-Level Supervision
```

Reviewer Question Addressed

> Do pseudo reports actually improve the local multimodal model?

---

10. Experiment 8: Full RASA Training

Aim

To evaluate the full report-anchored semantic alignment model.

Execution

代码块

```
python scripts/06_train_lcad_rasa.py \
  --config configs/train.yaml \
  --experiment full_lcad_rasa
```

Full Objective

代码块

```
L_total =
  λ_real × L_real_report
+ λ_pseudo × w_i × L_pseudo_report
+ λ_cls × L_binary_risk
+ λ_align × L_section_alignment
+ λ_cons × L_report_label_consistency
```

Corresponding Paper Section

代码块

```
Experiments 4.8 Report-Anchored Semantic Alignment
```

Reviewer Question Addressed

> Does the model perform real semantic alignment, or just multimodal concatenation?

---

11. Experiment 9: RASA Component Ablation

Aim

To prove that report-anchored semantic alignment contributes beyond ordinary fusion.

Execution

代码块

```
python scripts/08_run_ablation.py \
  --config configs/ablation.yaml \
  --manifest outputs/manifests/full_manifest_with_pseudo_reports.csv
```

Comparison

代码块

```
Simple fusion
Fusion + risk head
Fusion + section alignment
Full LCAD-RASA
```

Metrics

代码块

```
Report metrics on real-report cases
Clinical consistency
Section-source alignment score
Modality hallucination rate
Contradiction rate
```

Outputs

代码块

```
outputs/tables/rasa_component_ablation.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.9 Ablation of Report-Anchored Semantic Alignment
```

Reviewer Question Addressed

> Where is the semantic alignment? Does it improve performance?

---

12. Experiment 10: Agent Setting Ablation

Aim

To prove that LCAD is not a label-to-template generator.

Comparison

代码块

```
Label-only Agent
Modality-only Agent
Modality-plus-label Agent
```

Expected Evidence

The modality-plus-label Agent should improve:

代码块

```
Section completeness
Evidence consistency
Physician plausibility
Hallucination control
Impression consistency
```

relative to label-only generation.

Outputs

代码块

```
outputs/tables/agent_setting_ablation.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.10 Ablation of Agent Distillation Settings
```

Reviewer Question Addressed

> Are pseudo reports only reworded binary labels?

---

13. Experiment 11: QC Ablation

Aim

To test whether pseudo-report QC and confidence weighting are necessary.

Comparison

代码块

```
All pseudo reports
QC-passed pseudo reports only
QC-passed + confidence weighting
```

Metrics

代码块

```
Clinical contradiction rate
Hallucinated diagnosis rate
Report completeness
Cross-centre stability
Generated report quality
```

Outputs

```
outputs/tables/pseudo_report_qc_ablation.csv
```

Corresponding Paper Section

```
Experiments 4.11 Ablation of Pseudo-Report Quality Control
```

Reviewer Question Addressed

> Does QC matter, or are all pseudo reports equally useful?

---

14. Experiment 12: Modality Ablation

Aim

To quantify the contribution of each modality.

Comparison

```
OCT only
Colposcopy only
Instruction only
OCT + Colposcopy
OCT + Instruction
Colposcopy + Instruction
OCT + Colposcopy + Instruction
```

Metrics

```
Report generation metrics
Clinical consistency
Risk prediction performance
Section completeness
```

```
Hallucination rate
```

Outputs

代码块

```
outputs/tables/modality_ablation_summary.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.12 Modality Contribution Analysis
```

Reviewer Question Addressed

> Does the model really benefit from all three modalities?

---

15. Experiment 13: Modality Perturbation and Label Shuffle

Aim

To verify that the model uses multimodal evidence rather than memorising label-driven report templates.

Comparison

代码块

```
Normal input
Shuffle OCT
Shuffle colposcopy
Shuffle instruction
Mask OCT
Mask colposcopy
Mask instruction
Randomised label
```

Expected Result

If the model truly uses multimodal evidence, performance should degrade when relevant modalities are masked or shuffled.

Metrics

代码块

```
Risk AUC drop
Section consistency drop
Clinical contradiction increase
Hallucination increase
Physician plausibility decrease, if reviewed
```

Outputs

代码块

```
outputs/tables/modality_perturbation_summary.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.13 Evidence-Use Validation by Modality Perturbation
```

Reviewer Question Addressed

> Is the model actually using OCT, colposcopy and instruction evidence?

---

16. Experiment 14: Leave-One-Centre-Out Validation

Aim

To evaluate cross-centre generalisation.

Setting

Each centre is held out once as the test centre.

Metric Rules

If the held-out centre has real reports:

```
代码块Compute reference-based report metrics and clinical consistency.
```

If the held-out centre lacks real reports:

代码块

```
Do not compute BLEU/ROUGE as primary metrics.
Compute clinical consistency, section completeness, hallucination rate and physician-review readiness.
```

Outputs

代码块

```
outputs/tables/leave_one_center_out_summary.csv
```

Corresponding Paper Section

代码块

```
Experiments 4.14 Cross-Centre Generalisation
```

Reviewer Question Addressed

> Does the method generalise across hospital centres?

---

17. Experiment 15: Physician Blind Review

Aim

To provide clinical plausibility assessment for report-missing centres where no reference reports are available.

Execution

代码块

```
python scripts/10_export_physician_review_pack.py \
  --manifest outputs/manifests/full_manifest_with_pseudo_reports.csv \
  --generated_reports outputs/generated_reports/full_lcad_rasa \
  --output_dir outputs/physician_review \
```

```
    --cases_per_center 50
```

Review Dimensions

代码块

```
OCT evidence consistency
Colposcopy evidence consistency
Clinical context consistency
Impression plausibility
Recommendation usefulness
Overdiagnosis risk
Overall report quality
Free-text comments
```

Outputs

代码块

```
outputs/physician_review/review_cases.csv
outputs/physician_review/review_form_template.xlsx
outputs/physician_review/review_guide.md
```

Corresponding Paper Section

代码块

```
Experiments 4.15 Physician Blind Review
```

Reviewer Question Addressed

> Without real reports in four centres, how do you assess clinical plausibility?

---

18. Recommended Experimental Section Structure for the SCI Paper

4.1 Dataset and Multicentre Semantic Supervision Profiling

Report the number of centres, total cases, modality distribution, report-rich centre, report-missing centres, binary endpoint definition, missingness and data quality.

4.2 LCAD Calibration on the Report-Rich Centre

Report masking validation and label-only vs modality-only vs modality-plus-label results.

4.3 Pseudo-Report Generation and Quality Control

Report pseudo-report generation statistics, parse success rate, QC pass rate, label consistency rate, hallucination flags and confidence distribution.

4.4 Training Settings

Report real-only baseline, pseudo-augmented training, full LCAD-RASA, architecture, loss weights, pseudo-report weighting and section alignment implementation.

4.5 Main Performance Evaluation

Report reference-based metrics on real-report cases, clinical consistency on all centres, section completeness, hallucination rate and per-centre results.

4.6 Ablation Studies

Report Agent setting ablation, QC ablation, RASA component ablation, modality ablation and modality perturbation.

4.7 Cross-Centre Generalisation

Report leave-one-centre-out results, macro average, weighted average and separate metric interpretation for report-rich and report-missing held-out centres.

4.8 Physician Blind Review

Report sampling strategy, number of cases per centre, rating dimensions, inter-rater agreement if available and overall clinical plausibility.

---

19. Main Claims and Required Evidence

Claim	Required Experiment
The dataset has cross-centre semantic supervision imbalance	Exp 1
Agent pseudo reports are calibrated	Exp 3
LCAD is not label-template generation	Exp 3 and Exp 10
Pseudo-report QC is necessary	Exp 5 and Exp 11

Pseudo reports improve training	Exp 7
RASA improves over simple fusion	Exp 9
Model uses all three modalities	Exp 12 and Exp 13
Method generalises across centres	Exp 14
Report-missing centre outputs are clinically plausible	Exp 15

---

20. Main Paper Tables

Table 1: Dataset and Semantic Supervision Profile

Source:

代码块

```
centre_modality_summary.csv
report_availability_summary.csv
```

Table 2: Report-Rich Centre Masking Validation

Source:

代码块

```
report_rich_masking_validation_metrics.csv
```

Rows:

代码块

```
Label-only Agent
Modality-only Agent
Modality-plus-label Agent
```

Table 3: Pseudo-Report QC Summary

Source:

```
pseudo_report_quality_summary.csv
```

Table 4: Main Model Performance

Source:

代码块

```
eval_report_metrics.csv
eval_clinical_consistency.csv
eval_by_center.csv
```

Rows:

代码块

```
Real-only
Pseudo-augmented
Full LCAD-RASA
```

Table 5: Ablation Summary

Source:

代码块

```
agent_setting_ablation.csv
pseudo_report_qc_ablation.csv
rasa_component_ablation.csv
modality_perturbation_summary.csv
```

Table 6: Cross-Centre Generalisation

Source:

代码块

```
leave_one_center_out_summary.csv
```

---

21. Main Figures

Figure 1: Scientific Problem and Framework

Show 5 centres, one report-rich centre, four report-missing centres, semantic supervision imbalance, LCAD, RASA and structured report output.

Figure 2: LCAD Pseudo-Report Distillation

Show OCT evidence, colposcopy evidence, instruction evidence, binary label constraint, Agent teacher, structured pseudo report, QC and reliability weight.

Figure 3: RASA Architecture

Show OCT encoder to oct_findings, colposcopy encoder to colposcopy_findings, instruction encoder to clinical_context, fusion to impression/recommendation, risk head and report decoder.

Figure 4: Main Results

Show real-only vs pseudo-augmented vs full LCAD-RASA, clinical consistency, hallucination rate and section completeness.

Figure 5: Cross-Centre and Physician Review

Show leave-one-centre-out performance, physician rating distribution and example generated reports.

---

22. Wording for Experimental Results

Use cautious language:

代码块

```
Pseudo reports generated by LCAD were used as weak report-level supervision.
```

代码块

```
For report-missing centres, evaluation focused on clinical consistency,
hallucination analysis and physician review rather than reference-based text
similarity.
```

代码块

```
Report-rich masking validation was used to calibrate the pseudo-report
```

```
generation strategy.
```

Avoid:

代码块

```
The Agent generated ground-truth reports.
```

代码块

```
The model accurately diagnosed all patients.
```

代码块

```
BLEU/ROUGE proves clinical correctness.
```

---

23. Minimal Execution Order

For engineering execution:

代码块

```
bash run_minimal_pipeline.sh
```

For full paper experiments:

代码块

```
1. Data audit
2. Manifest construction
3. Modality evidence extraction
4. Report-rich masking validation
5. LCAD pseudo-report generation
6. Pseudo-report QC
7. Real-only training
8. Pseudo-augmented training
9. Full LCAD-RASA training
10. Main evaluation
11. Agent setting ablation
12. QC ablation
```

```
13. RASA component ablation
14. Modality ablation
15. Modality perturbation
16. Leave-one-centre-out validation
17. Physician review export
18. Paper table export
```

---

24. Final Experimental Story

The final experimental section should make the following argument:

> The dataset first demonstrates a real-world cross-centre semantic supervision imbalance. The report-rich centre masking validation then shows that LCAD can generate calibrated pseudo reports and that modality-plus-label distillation is superior to label-only generation. QC and confidence weighting reduce noisy supervision. When these pseudo reports are used together with real reports, the RASA model improves report quality, clinical consistency and section-level evidence alignment compared with real-only training and simple multimodal fusion. Modality ablation and perturbation confirm that the model uses all three modalities rather than relying on label templates. Leave-one-centre-out validation and physician blind review further support cross-centre robustness and clinical plausibility in report-missing centres.